

Automated Risk of Bias Assessment in Systematic Reviews Using Large Language Models: A Multi-Model Single-Agent Benchmark on 227 Randomised Controlled Trials

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Abstract

Background: Risk of bias (RoB) assessment is a mandatory but labour-intensive step in systematic review production. Large language models (LLMs) offer the potential to automate this task in a zero-shot setting, yet systematic benchmarks across multiple frontier models and inference strategies are lacking. **Methods:** Eight state-of-the-art LLMs (Gemini 2.5 Pro, Gemini 2.5 Flash, GPT-5.4, Claude Haiku 4.5, DeepSeek v4-Flash, Grok 4.3, Qwen 3.7-Plus, and GLM-5) were evaluated on zero-shot, single-agent RoB 1.0 assessment of 227 randomised controlled trials drawn from 127 Cochrane systematic reviews published in 2024. Models were accessed via the OpenRouter API in June 2026 using a fixed prompt and provider-default generation parameters. Ground-truth annotations were taken from the published Cochrane consensus judgements across seven methodological domains (1,589 cells total). Performance was assessed using accuracy, Macro-F1, and Gwet's AC1,

with study-clustered bootstrap 95% confidence intervals. Two models (Gemini 2.5 Flash and DeepSeek v4-Flash) were additionally evaluated under a single-call inference mode for comparison with the per-domain strategy.

Results: Overall accuracy ranged from 57.6% to 62.5% across models, a spread smaller than the width of individual 95% confidence intervals; Macro-F1 confidence intervals overlapped across all eight models. AC1 provided the clearest separation: Claude Haiku 4.5 (AC1 = 0.485; 95% CI [0.443, 0.526]) was distinguishable from Gemini 2.5 Pro ([0.344, 0.420]) and GPT-5.4 ([0.349, 0.426]). Unclear Risk was the most consistently under-detected class (recall < 0.20 in three of seven domains across most models). The oracle accuracy of 85.8%—23 percentage points above the best individual model—confirmed substantial non-redundancy among models. The comparison of inference modes yielded small, largely non-significant differences: ten of twelve Δ metrics had confidence intervals spanning zero, and D7 (Other Bias)—the domain hypothesised to benefit most from joint assessment—showed negligible change under either mode.

Conclusions: Frontier LLMs can perform zero-shot RoB 1.0 assessment at accuracy levels that approach the noise floor of the human reference standard, but no single model dominates across all metrics and no consistent advantage was found for single-call over per-domain inference. The primary practical distinction between inference modes is computational: single-call reduces inference calls by $7\times$ without a reliable accuracy trade-off. All analyses are exploratory; results should be interpreted as directional benchmarks rather than definitive rankings.

Keywords: risk of bias; systematic review; large language models; natural language processing; Cochrane; evidence synthesis; automation

1 Introduction

Systematic reviews and meta-analyses occupy the apex of the evidence hierarchy in evidence-based medicine [1]. Their validity depends critically on the quality of the primary studies included: a meta-analysis that aggregates trials with uncontrolled confounding or selective outcome reporting inherits—and can amplify—those methodological flaws. The Cochrane Risk of Bias 1.0 tool (RoB 1.0) [2] was developed precisely to make this quality appraisal explicit and reproducible. By requiring reviewers to judge each included trial across seven pre-specified methodological domains, RoB 1.0 transforms a subjective quality impression into a structured, auditable record.

Despite this formalisation, RoB assessment remains a major bottleneck in systematic review production. Each trial must be read by two independent reviewers, who compare judgements and resolve disagreements through discussion or adjudication. For a review covering dozens of trials, this process routinely requires tens to hundreds of person-hours. Inter-rater reliability, even among trained reviewers, is moderate at best [3, 4]—a consequence of genuine ambiguity in reporting, domain-specific expertise requirements, and the epistemic nature of the *Unclear Risk* category, which denotes insufficient information rather than a graded risk level. These limitations have motivated sustained interest in automating, or at least augmenting, the RoB assessment workflow.

Early automation efforts relied on rule-based systems and supervised machine learning trained on sentence-level text fragments [5]. These approaches demonstrated that natural language processing could identify methodological descriptions in trial reports with above-chance accuracy, but they were constrained by limited contextual understanding, sensitivity to reporting style, and the need for substantial labelled training data for each domain.

Large language models (LLMs) trained on diverse corpora with instruction-following capabilities represent a qualitatively different paradigm [6, 7]. Without task-specific fine-tuning, general-purpose LLMs can be prompted to read a full trial

76 report, locate domain-relevant passages, and produce structured judgements with
77 free-text justifications—mimicking the expert reviewer workflow. A prompt template
78 can address all seven RoB 1.0 domains either sequentially—one domain per inference
79 call—or simultaneously in a single call, and whether this choice affects assessment
80 quality is an open empirical question. As frontier LLMs have grown more capable
81 and their inference costs have fallen, evaluating their effectiveness for RoB assessment
82 has become both feasible and practically important.

83 The present study provides a systematic empirical comparison of eight state-of-the-
84 art LLMs on zero-shot, single-agent RoB 1.0 assessment applied to 227 randomised
85 controlled trials drawn from 127 Cochrane systematic reviews published in 2024.
86 All models were evaluated on a common benchmark under identical conditions:
87 the same prompt, the same fully text-based study representations produced by
88 a standardised PDF-to-Markdown pipeline, and the same ground-truth human
89 annotations. We report accuracy, Macro-F1, and Gwet’s AC1—metrics selected to
90 characterise both aggregate performance and class-level sensitivity without imposing
91 an ordinal interpretation on the three nominal RoB categories. Stratified analyses by
92 intervention type and clinical domain identify sources of systematic variation across
93 the benchmark. Additionally, we compare two inference strategies—per-domain and
94 single-call—using Gemini 2.5 Flash and DeepSeek v4-Flash, the two models evaluated
95 under both conditions, to assess whether querying all seven domains simultaneously
96 affects performance relative to sequential per-domain calls.

97 **2 Methods**

98 **2.1 Dataset**

99 The ground-truth dataset was constructed from Cochrane systematic reviews pub-
100 lished in 2024. Of the 149 reviews published that year, 127 were accessible through
101 our institutional subscription; from each accessible review, up to three included
102 randomised controlled trials (RCTs) were selected based on full-text PDF availability:

all included RCTs for which a PDF was retrievable were included, up to a maximum of three per review, yielding a final corpus of **227 RCTs** drawn from 127 distinct Cochrane reviews.

Risk of bias annotations were obtained directly from the published Cochrane review reports and reflect the consensus judgements produced by the original review teams following standard Cochrane procedures (independent assessment by at least two trained reviewers, with disagreements resolved by discussion or arbitration). These annotations were not re-produced or independently adjudicated by the present study; they represent the published Cochrane consensus, which is the reference standard used by the field. Although the revised Risk of Bias 2.0 tool (RoB 2; Sterne et al. 8) is now recommended for new systematic reviews, it reorganises the domain structure and changes the rating scale, making it incompatible with the existing RoB 1.0 corpus. RoB 1.0 remains the basis for the large majority of existing systematic review annotations — including all 127 reviews sampled here and most of the large labelled datasets used for training and evaluation of automated systems — so this study’s focus on RoB 1.0 is both corpus-driven and practically motivated. Each study was annotated using the Cochrane Risk of Bias 1.0 (RoB 1.0) tool [2], which covers seven methodological domains: D1 (random sequence generation), D2 (allocation concealment), D3 (blinding of participants and personnel), D4 (blinding of outcome assessment), D5 (incomplete outcome data), D6 (selective outcome reporting), and D7 (other sources of bias). Each domain received one of three judgements—*Low Risk*, *High Risk*, or *Unclear Risk*—yielding 1,589 annotated cells (227×7).

Class prevalence varied substantially across domains (Table 1). Low Risk was the largest class in six of seven domains, and the majority class ($>50\%$) in four of them (D1, D5, D6, D7); only D3 (blinding of participants and personnel) had High Risk as its largest class (45%), reflecting the predominance of unblinded pharmacological trials in the corpus. At the skewed extreme, D1 had a 69% Low Risk majority and only 5 High Risk cases (2%); D5 and D7 were similarly imbalanced at 63% and 61%

Low Risk, respectively. D4 (blinding of outcome assessors) was the most balanced domain overall (44% Low Risk, 28% High Risk, 29% Unclear Risk), closely followed by D2, in which Low Risk and Unclear Risk were nearly equally represented (50% vs. 46%).

Table 1: Ground-truth class distribution per RoB domain ($n = 227$ studies per domain). LR = Low Risk; HR = High Risk; UR = Unclear Risk.

Domain	LR	HR	UR	Majority class	Majority (%)
D1	157	5	65	Low Risk	69.2
D2	113	10	104	Low Risk	49.8
D3	74	103	50	High Risk	45.4
D4	99	63	65	Low Risk	43.6
D5	143	39	45	Low Risk	63.0
D6	125	25	77	Low Risk	55.1
D7	139	31	57	Low Risk	61.2
Total	850 (53.5%)	276 (17.4%)	463 (29.1%)	Low Risk	53.5

Intervention type. Each study was assigned a single intervention category: drug/medication ($n = 104$), non-pharmacological intervention ($n = 75$), procedure ($n = 30$), or medical device ($n = 18$).

Clinical domain. Studies were tagged with one or more of 24 Cochrane Review Group categories (multi-label; 120 of 227 studies belonged to more than one group). The most frequent categories were Child Health ($n = 72$), Heart & Circulation ($n = 45$), and Kidney Disease ($n = 31$); the least frequent were Skin Disorders ($n = 3$), Effective Practice & Health Systems ($n = 4$), Rheumatology ($n = 5$), and Wounds ($n = 5$). For stratified analyses, each multi-domain study was included in every applicable group.

2.2 Document Preparation Pipeline

Full-text study reports (PDF) were processed through a two-stage pipeline designed to produce fully text-based study representations that can be consumed by any large language model without multimodal (image) input.

Stage 1 — PDF-to-Markdown conversion. Each PDF was converted

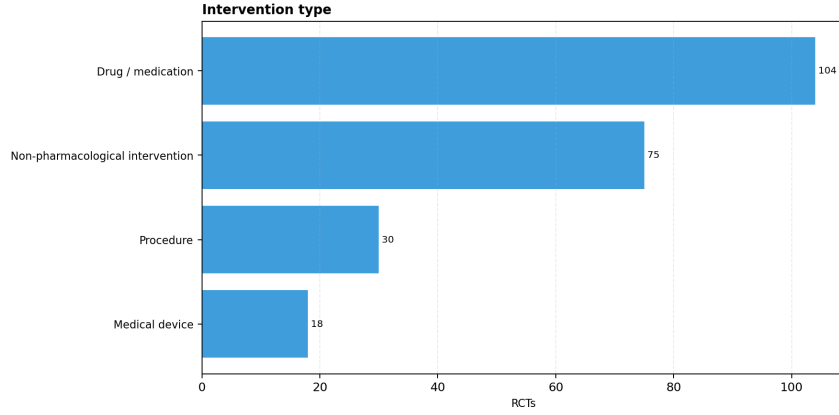


Figure 1: Distribution of the 227 RCTs by intervention type.

151 to structured Markdown using MinerU [9], which preserves headings, tables, and
 152 inline text while replacing embedded figures with placeholder tokens of the form
 153 [Figure *k*].

154 **Stage 2 — Figure description.** Each figure placeholder was resolved by a vision-
 155 language model (VLM) that received the corresponding page image and generated a
 156 textual description, which was then inserted in place of the placeholder. After Stage 2,
 157 the document was a self-contained Markdown file in which all information—including
 158 the content of charts, CONSORT flow diagrams, Kaplan–Meier curves, and forest
 159 plots—was represented as plain text. This design eliminates the need to pass images
 160 alongside the prompt during RoB assessment, enabling evaluation of text-only models
 161 and ensuring a consistent input format across all eight models tested.

162 2.3 Risk of Bias Assessment

163 All models were evaluated in a zero-shot, single-agent setting: each processed
 164 Markdown document was submitted to the model with a fixed assessment prompt
 165 (reproduced in full in Supporting Information S1), which specified the domain
 166 criteria, the three admissible judgements (Low Risk, High Risk, Unclear Risk), and
 167 the required JSON output format. No few-shot examples were provided, and the
 168 prompt was used without modification across all models.

169 **Inference modes.** We evaluated two inference strategies that differ in how
 170 the seven RoB domains are requested from the model. In the *per-domain* mode,

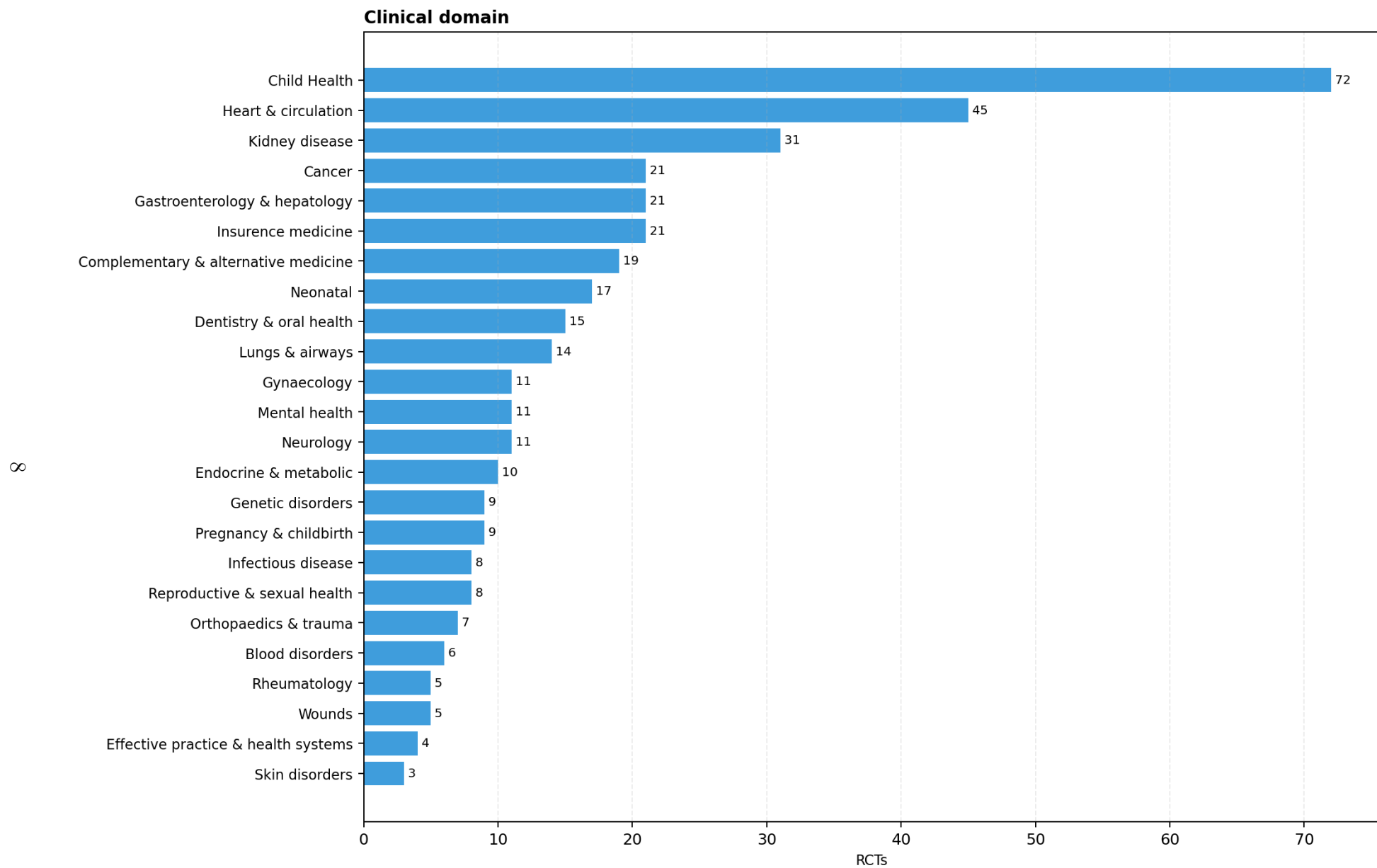


Figure 2: Distribution of the 227 RCTs by clinical domain (Cochrane Review Group categories, multi-label; total exceeds 227).

171 seven sequential inference calls are made per study: each call supplies the full
172 Markdown text together with a single domain identifier (e.g. `Domain: D1`), and the
173 model returns a JSON object with a judgement and justification for that domain
174 only. In the *single-call* mode, a single inference call per study is made: the model
175 receives the full Markdown text and is instructed to return a JSON array containing
176 judgements and justifications for all seven domains simultaneously. All eight models
177 were evaluated under the per-domain mode. Gemini 2.5 Flash and DeepSeek v4-
178 Flash were additionally evaluated under the single-call mode on the same 227 studies,
179 enabling a controlled comparison of the two inference strategies while holding model,
180 prompt content, and corpus constant.

181 2.4 Language Models

182 Eight state-of-the-art LLMs were evaluated (Table 2). All models were accessed via
183 the OpenRouter API in June 2026 using the identifier strings listed in Table 2. No
184 fine-tuning or prompt adaptation specific to individual models was performed. All
185 inference calls used provider-default sampling parameters (temperature and other
186 generation settings at the API defaults); each study received a single inference
187 call per condition (one call per domain in per-domain mode; one call for all seven
188 domains in single-call mode), with no repeated sampling. This design was intentional:
189 our aim is to benchmark models as they are available to practitioners, not under
190 artificial conditions. Manipulating generation parameters such as temperature would
191 (i) require a separate optimisation procedure for each model, introducing a confound
192 between prompt design and generation settings, and (ii) produce heterogeneous
193 evaluation conditions given that different model families exhibit varying sensitivity to
194 the same temperature value. Strategies such as majority voting over repeated draws
195 or ensembling outputs across runs — which can improve reliability — represent
196 post-hoc optimisation beyond the scope of a zero-shot benchmark; their evaluation is
197 left to future work. The OpenRouter identifier strings in Table 2 represent the model
198 state routed at the time of inference; providers may silently update the underlying

weights associated with a fixed identifier string, so results reported here reflect the model checkpoints active during the June 2026 inference window.

Table 2: Language models evaluated. All models accessed via the OpenRouter API in June 2026 using the model identifier strings shown; zero-shot single-agent mode throughout. Context length: maximum tokens accepted per call as reported by OpenRouter at access time. Release date: date the model became available on OpenRouter. Exact checkpoint versions are opaque at the provider level; the identifiers below are the strings passed to the API and represent the model state at the time of inference.

Model	OpenRouter identifier	Context	Release	Studies
Gemini 2.5 Pro	google/gemini-2.5-pro	1M	Jun 2025	227
Gemini 2.5 Flash	google/gemini-2.5-flash	1M	Jun 2025	227
GPT-5.4	openai/gpt-5.4	1M	Mar 2026	227
Claude Haiku 4.5	anthropic/claude-haiku-4.5	200K	Oct 2025	227
DeepSeek v4-Flash	deepseek/deepseek-v4-flash	1M	Apr 2026	227
Grok 4.3	x-ai/grok-4.3	1M	Apr 2026	227
Qwen 3.7-Plus	qwen/qwen3.7-plus	1M	May 2026	227
GLM-5	z-ai/glm-5	203K	Feb 2026	227

2.5 Evaluation Metrics

Overall accuracy. Proportion of cells in which the model’s judgement matched the ground-truth label, computed across all 1,589 study–domain pairs.

Per-class F1. For each class $k \in \{\text{Low Risk, High Risk, Unclear Risk}\}$, the F1 score is the harmonic mean of precision and recall:

$$F1_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}, \quad (1)$$

computed across all study–domain cells. Per-class F1 reveals class-specific failure modes independently of overall performance.

Macro-F1. Unweighted mean of the three per-class F1 scores: $\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K F1_k$ with $K = 3$. Macro-F1 assigns equal weight to each category regardless of prevalence, penalising models that achieve high accuracy by concentrating predictions on majority classes.

Gwet’s AC1. A chance-corrected agreement coefficient that is robust to the

prevalence paradox [10, 11]: when marginal distributions are highly skewed, classical chance-corrected coefficients approach zero even when observed agreement is substantial. AC1 is defined as

$$\text{AC1} = \frac{p_o - p_e}{1 - p_e}, \quad p_e = \frac{1}{K-1} \sum_{k=1}^K \pi_k(1 - \pi_k), \quad (2)$$

where p_o is the observed proportion of agreement, $K = 3$ is the number of categories, and $\pi_k = (\hat{p}_{k,\text{model}} + \hat{p}_{k,\text{gt}})/2$ is the mean marginal proportion for category k , with $\hat{p}_{k,\text{model}}$ and $\hat{p}_{k,\text{gt}}$ the marginal proportions of class k in the model predictions and ground truth, respectively. All metrics—including AC1—are computed globally by pooling all 1,589 study–domain cells; domain-level values are not aggregated. Given the pronounced class imbalance documented in Table 1—six of seven domains have Low Risk as the majority class—AC1 is more appropriate than Cohen’s κ as the primary agreement metric for this task, since κ approaches zero under heavy prevalence skew even when observed agreement is substantial (the prevalence paradox; Feinstein and Cicchetti 11). Importantly, the ground-truth annotations used here reflect published Cochrane consensus judgements rather than a second independent annotator; AC1 therefore quantifies model-to-published-standard agreement, not inter-rater reliability in the classical sense. Human inter-rater reliability for RoB 1.0 is moderate at best: even among formally trained reviewers applying the same protocol, Hartling et al. [12] reported $\kappa = 0.79$ for D1 but only $\kappa = 0.24$ – 0.37 for the remaining domains, with agreement between independent consensus pairs falling further to $\kappa = 0.05$ – 0.09 in several domains. Agreement also varies substantially between distinct Cochrane review teams evaluating the same trials [3]. These findings contextualise model performance against a noisy reference standard rather than a true human ceiling, and motivate domain-level rather than aggregate interpretation of results.

Oracle upper bound. For each study–domain cell, the oracle is correct if at least one of the eight models matched the ground truth. The oracle accuracy quantifies the theoretical ceiling achievable by a hypothetical perfect model-selection

strategy and characterises the degree of complementarity among models.

Stratified analyses. All primary metrics were recomputed within subgroups defined by intervention type (four categories) and clinical domain (24 Cochrane Review Groups, multi-label). For clinical domain, studies were included in every group to which they belonged, so group sample sizes sum to more than 227.

Statistical inference. All analyses are descriptive and exploratory. Confidence intervals are reported to characterise estimation uncertainty and facilitate comparison across models and conditions; they are not interpreted as the outcome of pre-specified confirmatory hypothesis tests. No familywise error rate correction was applied: the number of comparisons across models, domains, classes, and inference modes is large and the outcomes are correlated, making conventional multiplicity corrections either too conservative (Bonferroni) or difficult to calibrate without a pre-specified primary endpoint. Where 95% bootstrap CIs exclude zero, this is noted descriptively (“CI excludes zero”) and should be interpreted as an unadjusted signal warranting further investigation, not as a confirmatory claim.

3 Results

3.1 Overall Performance

Table 3 reports accuracy, Macro-F1, per-class F1, and Gwet’s AC1 for all eight models across the 1,589 annotated cells, together with study-clustered bootstrap 95% confidence intervals (5,000 iterations, resampling the 227 studies with replacement to preserve within-study correlation across the seven domains). Overall accuracy ranged from 57.6% (Gemini 2.5 Pro) to 62.5% (Claude Haiku 4.5), a spread of less than five percentage points; the 95% CI for each model spans approximately 5–6 pp, so all pairwise accuracy differences fall within sampling uncertainty and should be treated as directional. Macro-F1 confidence intervals similarly overlap across all eight models, precluding reliable rank ordering based on this metric alone. By contrast, Gemini 2.5 Flash and DeepSeek v4-Flash tied for the highest Macro-F1

(0.558) while Claude Haiku 4.5—the accuracy leader—ranked sixth (0.541). The per-class breakdown reveals that the Macro-F1 gap is driven primarily by High Risk and Unclear Risk detection: Claude Haiku 4.5 achieved the highest Low Risk F1 (0.743) but the lowest High Risk F1 (0.469) and Unclear Risk F1 (0.413) among all models. Among the three metrics, Gwet’s AC1 shows the clearest separation at the extremes: the 95% CI for Claude Haiku 4.5 ($AC1 = 0.485$; $CI [0.443, 0.526]$) does not overlap with those of Gemini 2.5 Pro ($[0.344, 0.420]$) or GPT-5.4 ($[0.349, 0.426]$), indicating that Haiku’s higher chance-corrected agreement is not attributable to sampling variation. The oracle upper bound reached **85.8%**, representing a 23.3 percentage-point gap above the best individual model.

Metric divergence. The discrepancy between accuracy and Macro-F1 rankings reflects differential handling of the class imbalance documented in Table 1. The per-class F1 columns in Table 3 make the trade-off explicit: models with higher Low Risk F1 tend to achieve higher accuracy but lower High Risk and Unclear Risk F1, since Low Risk is the majority class in six of seven domains. Macro-F1 penalises this behaviour by weighting each class equally, shifting the ranking toward models with more balanced per-class detection.

3.2 Performance by RoB Domain

Domain-level accuracy (Figure 3; Table SS1 in Supporting Information) revealed two tiers of difficulty. **D1** (sequence generation) and **D2** (allocation concealment) were the most accurately assessed domains, with mean accuracy of 75.3% and 72.7%, respectively; all models exceeded 70% accuracy on D1. **D3** (blinding of participants) showed high variance across models (49.8%–72.2%), with Gemini 2.5 Flash and Claude Haiku 4.5 outperforming others by a margin of more than 5 percentage points.

D7 (other sources of bias) was the most challenging domain for all models, with accuracy ranging from 24.2% (Gemini 2.5 Pro) to 55.5% (Qwen 3.7-Plus)—a spread of 31 percentage points within a single domain. Notably, models at opposite

Table 3: Overall performance on RoB 1.0 assessment ($n = 227$ studies; 1,589 cells). Models ranked by accuracy. F1-LR, F1-HR, F1-UR: F1 for Low Risk, High Risk, and Unclear Risk computed globally across all study–domain cells. Best value in each column shown in bold. 95% confidence intervals (study-clustered bootstrap, 5,000 iterations, resampling 227 studies with replacement) are shown in *brackets* below each point estimate; all analyses are exploratory (see Methods, Statistical Inference).

Model	Accuracy (%)	Macro-F1	F1-LR	F1-HR	F1-UR	Gwet’s AC1
Claude Haiku 4.5	62.5 [59.8, 65.2]	0.541 [0.511, 0.569]	0.743 [0.718, 0.767]	0.469 [0.412, 0.521]	0.413 [0.361, 0.463]	0.485 [0.443, 0.526]
GLM-5	61.1 [58.7, 63.5]	0.557 [0.529, 0.582]	0.709 [0.682, 0.735]	0.441 [0.385, 0.494]	0.520 [0.479, 0.561]	0.449 [0.411, 0.485]
Qwen 3.7-Plus	60.9 [58.3, 63.4]	0.525 [0.496, 0.552]	0.718 [0.691, 0.743]	0.356 [0.293, 0.413]	0.503 [0.459, 0.543]	0.457 [0.417, 0.495]
Gemini 2.5 Flash	60.7 [58.2, 63.1]	0.558 [0.533, 0.582]	0.718 [0.690, 0.744]	0.474 [0.424, 0.516]	0.484 [0.443, 0.526]	0.437 [0.399, 0.475]
DeepSeek v4-Flash	60.3 [57.7, 62.9]	0.558 [0.531, 0.585]	0.704 [0.676, 0.730]	0.480 [0.429, 0.530]	0.489 [0.447, 0.530]	0.433 [0.394, 0.472]
Grok 4.3	60.1 [57.8, 62.5]	0.549 [0.523, 0.574]	0.696 [0.670, 0.722]	0.432 [0.376, 0.486]	0.518 [0.478, 0.556]	0.433 [0.397, 0.471]
GPT-5.4	57.7 [55.1, 60.3]	0.546 [0.519, 0.572]	0.668 [0.636, 0.698]	0.464 [0.410, 0.511]	0.506 [0.468, 0.543]	0.387 [0.349, 0.426]
Gemini 2.5 Pro	57.6 [55.1, 60.0]	0.543 [0.517, 0.569]	0.674 [0.643, 0.702]	0.455 [0.407, 0.501]	0.502 [0.458, 0.542]	0.383 [0.344, 0.420]
<i>Oracle</i>	<i>85.8</i>	—	—	—	—	—

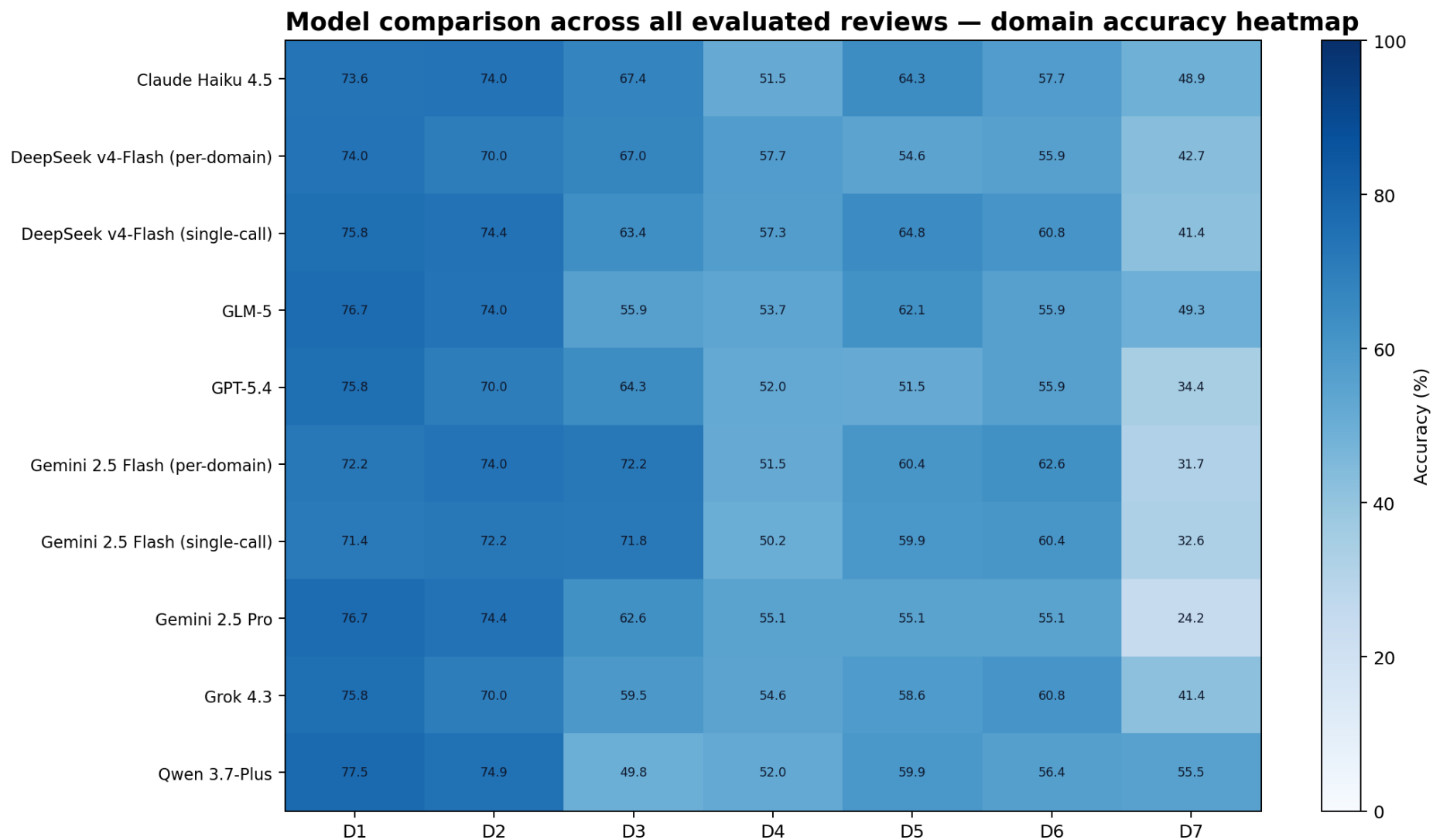


Figure 3: Per-domain accuracy (%) for each model. Colour scale: 0–100%.

ends of the D7 accuracy range adopted diametrically opposed prediction strategies: Gemini 2.5 Pro assigned High Risk to over 93% of D7 cells (recall 0.94 for High Risk; recall 0.18 for Low Risk), whereas Qwen 3.7-Plus assigned Low Risk to 88% of D7 cells (recall 0.88 for Low Risk; recall 0.06 for High Risk). Both strategies yielded similarly poor Macro-F1 in D7 (≤ 0.37 for both), demonstrating that high accuracy on a single class does not translate to overall competence.

The oracle accuracy per domain was highest for D2 (91.2%) and lowest for D4 (78.0%), indicating that the 226 cells where no model was correct were not uniformly distributed across domains.

3.3 Per-class Analysis

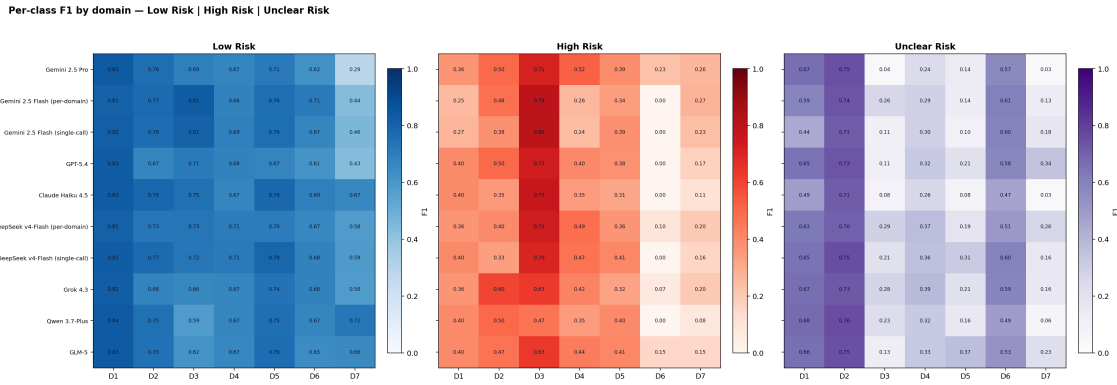


Figure 4: F1 scores for Low Risk (blue), High Risk (red), and Unclear Risk (purple) by domain and model. Colour intensity proportional to F1 (0–1).

Figure 4 disaggregates Macro-F1 into its three constituent classes.

Low Risk detection was strong and consistent across models in D1 and D2 ($F1 > 0.75$ for all models). Performance deteriorated in D7, where most models reported Low Risk F1 below 0.60; Gemini 2.5 Pro collapsed to $F1 = 0.29$ in D7 as a consequence of its High Risk-biased prediction strategy described above.

High Risk detection was reliable only in D3 ($F1: 0.47\text{--}0.79$), the single domain in which High Risk cases outnumbered Low Risk cases in the ground truth (103 vs. 74; Table 1). In D6 (selective outcome reporting), four models reported High Risk $F1 = 0.00$ (Gemini 2.5 Flash, GPT-5.4, Claude Haiku 4.5, and Qwen 3.7-Plus), corresponding to zero true-positive High Risk predictions across all 227 studies in

that domain. High Risk is the least prevalent class in D6 (25 of 227 cases, 11%; Table 1), and these models predicted fewer than three High Risk cases in D6 each, systematically defaulting to Low Risk or Unclear Risk in the absence of explicit evidence of selective reporting.

Unclear Risk was the most consistently under-detected class. Mean recall across all models fell below 0.20 in D3, D5, and D7. The worst single result was Claude Haiku 4.5 in D5: recall = 0.04, meaning the model correctly identified only 2 of the 45 Unclear Risk cases in D5. The most accurately detected Unclear Risk domain was D2, where all models achieved $F1 > 0.70$, consistent with the near-equal prevalence of Low Risk and Unclear Risk judgements in that domain (113 vs. 104).

3.4 Stratification by Intervention Type

Mean Macro-F1 across all models ranked intervention types as follows: non-pharmacological intervention (0.572, $n = 75$ studies), drug/medication (0.538, $n = 104$), medical device (0.531, $n = 18$), and procedure (0.510, $n = 30$). The 6.2-point gap between non-pharmacological interventions and procedures was consistent across most models; no single model reversed this ordering. Individual model rankings within each intervention type are provided in Table SS3.

3.5 Stratification by Clinical Domain

Among clinical domains with at least five studies, mean Macro-F1 across models ranged from 0.425 (Infectious Disease, $n = 8$) to 0.680 (Orthopaedics & Trauma, $n = 7$)—a 25.5-point spread (Figure 5). The four highest-performing domains were Orthopaedics & Trauma (0.680), Endocrine & Metabolic (0.638, $n = 10$), Rheumatology (0.637, $n = 5$), and Wounds (0.628, $n = 5$). The four lowest-performing domains were Infectious Disease (0.425), Reproductive & Sexual Health (0.437, $n = 8$), Genetic Disorders (0.473, $n = 9$), and Mental Health (0.481, $n = 11$). Complete results for all 24 Cochrane Review Groups are provided in Table SS4.

These results are hypothesis-generating rather than confirmatory. Group sizes of

5–11 studies at the extremes preclude reliable rank-ordering across domains, and multi-label tagging makes groups non-exclusive. The pattern is nonetheless qualitatively consistent with an interpretable explanation: domains characterised by standardised reporting (orthopaedic interventions and metabolic trials typically follow structured protocols with explicit randomisation and blinding procedures) show higher Macro-F1, while domains with heterogeneous or complex study designs (Infectious Disease, Reproductive & Sexual Health) present harder RoB characterisation challenges for models trained on general text. These observations motivate targeted future evaluation on larger domain-specific corpora.

3.6 Oracle Analysis

The oracle accuracy of 85.8% (1,363 of 1,589 cells) establishes a theoretical upper bound for any strategy that selects among the eight evaluated models; it is not achievable in practice, since it requires prior knowledge of which model is correct for each cell. Of the 226 cells where no model was correct, 50 (22.1%) occurred in D4 and 44 (19.5%) in D5—the two domains with the most balanced three-class distributions and, correspondingly, the lowest oracle ceilings (78.0% and 80.2%, respectively). Pairwise complementarity analysis (Supporting Information S) showed that for any two models, between 7.4% and 17.4% of cells were correctable by substituting one model for the other, confirming substantial non-redundant coverage across the ensemble.

3.7 Inference Mode Comparison

Table 4 compares per-domain and single-call inference modes for the two models evaluated under both conditions.

The overall picture from Table 4 is one of small differences with wide uncertainty: of 12 Δ values across both models and six metrics, ten have 95% CIs that include zero. The two whose CIs exclude zero are a drop in Unclear Risk F1 for Gemini 2.5 Flash (−0.044; CI [−0.079, −0.009], †) and a gain in Low Risk F1 for DeepSeek v4-Flash

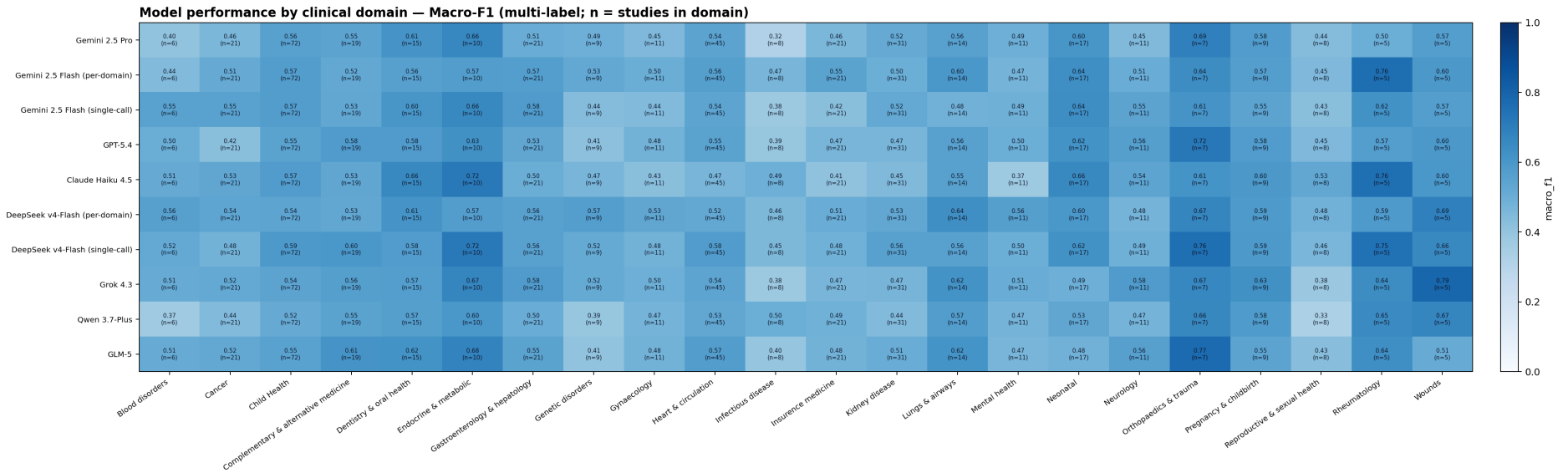


Figure 5: Macro-F1 by clinical domain and model (domains with ≥ 5 studies; n = number of studies in each domain, multi-label). Colour scale: 0–1.

Table 4: Per-domain vs. single-call inference mode comparison for Gemini 2.5 Flash and DeepSeek v4-Flash ($n = 227$ studies; 1,589 cells each). F1 scores are computed globally across all cells (identical methodology to Table 3); per-domain values reproduce the corresponding entries in Table 3. Δ : single-call minus per-domain; 95% paired bootstrap CI (study-clustered, 5,000 iterations) shown in brackets. $\dagger$: CI excludes zero (unadjusted for multiplicity; all analyses exploratory).

Model	Accuracy (%)	Macro-F1	F1-LR	F1-HR	F1-UR
<i>Gemini 2.5 Flash</i>					
Per-domain	60.7	0.558	0.718	0.474	0.484
Single-call	59.8	0.548	0.715	0.490	0.440
Δ	−0.9	−0.010	−0.003	+0.016	−0.044 [†]
	[−2.7, +1.0]	[−0.030, +0.011]	[−0.019, +0.014]	[−0.013, +0.051]	[−0.079, −0.009]
<i>DeepSeek v4-Flash</i>					
Per-domain	60.3	0.558	0.704	0.480	0.489
Single-call	62.6	0.576	0.730	0.475	0.523
Δ	+2.3	+0.018	+0.026 [†]	−0.005	+0.034
	[−0.1, +4.7]	[−0.007, +0.044]	[+0.004, +0.047]	[−0.049, +0.041]	[−0.003, +0.072]

(+0.026; [+0.004, +0.047], [†]); both are class-specific and do not translate into differences in accuracy, Macro-F1, or AC1, all of which remain within their respective uncertainty intervals. The accuracy differences (−0.9 pp for Flash, +2.3 pp for DeepSeek), while pointing in opposite directions, are each compatible with zero change given the bootstrap CIs ([−2.7, +1.0] and [−0.1, +4.7] respectively).

Domain-level results reinforce this picture. D7 (Other Bias), the domain for which joint assessment was hypothesised to confer the greatest benefit through shared contextual reasoning, showed negligible differences under both modes (± 1 pp)— the most direct evidence that collapsing seven domain assessments into a single call does not meaningfully improve the handling of the hardest domain. The opposing D3 pattern (Flash: −0.4 pp; DeepSeek: −3.6 pp) and the concentration of DeepSeek gains in D5 (+10.2 pp) and D2 (+4.4 pp) suggest that whatever domain-level shifts occur are idiosyncratic to each model rather than reflecting a systematic benefit of joint assessment.

Single-call outputs were systematically shorter (DeepSeek: 400→214 chars; Gemini 2.5 Flash: 530→354 chars), with a corresponding decrease in hedging expressions (e.g., “unclear”, “not reported”, “cannot determine”) in free-text justifications, con-

386 sistent with a more compressed generation regime under the single-call prompt.
387 The practical implications of this output compression appear limited and model-
388 dependent: it did not translate into a consistent performance benefit or detriment
389 across either model at the aggregate level.

390 4 Conclusions

391 Eight state-of-the-art LLMs assessed RoB 1.0 for 227 randomised controlled trials in
392 a zero-shot, single-agent setting. Overall accuracy ranged narrowly from 57.6% to
393 62.5%, but Macro-F1 and Gwet’s AC1 revealed meaningful differences in how models
394 handled the three nominal classes. Models with the highest accuracy achieved it
395 partly by concentrating predictions on majority-class labels, a strategy that inflates
396 accuracy while suppressing detection of High Risk and Unclear Risk judgements—the
397 very categories most consequential for evidence appraisal. Macro-F1 and Gwet’s AC1
398 are therefore more informative primary metrics than accuracy for this task.

399 *Unclear Risk* was the most consistently under-detected class across all models
400 and domains, with recall below 0.20 in three of seven domains. This systematic
401 failure reflects the epistemic character of the category—it requires the model to
402 recognise that key methodological information is absent, a more complex inference
403 than identifying a positive or negative methodological signal in the text.

404 Domain-level and stratification analyses revealed structured heterogeneity in
405 performance. Sequence generation (D1) and allocation concealment (D2) were
406 reliably assessed across all models, whereas other sources of bias (D7) showed the
407 greatest inter-model accuracy spread and blinding of outcome assessors (D4) had
408 the lowest oracle ceiling (78.0%). Clinical domains with clear, domain-specific
409 methodological criteria (Orthopaedics & Trauma, Endocrine & Metabolic) supported
410 higher Macro-F1 than domains with more heterogeneous study designs (Infectious
411 Disease, Reproductive & Sexual Health).

412 The oracle accuracy of 85.8%—23 percentage points above the best individual
413 model—confirms that the eight models make substantially non-redundant errors,

414 establishing a firm empirical ceiling for any multi-model selection strategy.

415 The comparison between per-domain and single-call inference modes (Section 3.7)
416 produced small differences that do not collectively support a performance advantage
417 for either strategy. Ten of twelve Δ values across both models and six metrics
418 had 95% CIs spanning zero; the two exceptions were class-specific effects of limited
419 practical magnitude (Unclear Risk F1 for Flash: -0.044 ; Low Risk F1 for DeepSeek:
420 $+0.026$) that did not propagate to aggregate metrics. Most notably, D7 (Other Bias)—
421 the domain for which joint contextual assessment was hypothesised to be most
422 beneficial—showed negligible differences under both modes (± 1 pp), indicating that
423 its difficulty is intrinsic to the vagueness of the criterion rather than a consequence
424 of fragmented inference calls. Taken together, the preliminary evidence from this
425 two-model comparison does not identify a meaningful performance rationale for
426 preferring single-call over per-domain assessment; the primary practical distinction
427 between the two modes is computational (single-call reduces inference calls by $7\times$)
428 rather than one of accuracy.

429 Several limitations should be noted. First, the ground-truth annotations derive
430 from published Cochrane review reports and were not independently re-annotated for
431 this study. The Cochrane annotations represent the consensus output of a structured
432 multi-reviewer process (independent assessment by at least two trained reviewers,
433 with disagreements resolved by discussion or arbitration), which is the reference
434 standard used operationally by the field and the most appropriate benchmark for
435 evaluating a tool intended to assist real-world systematic review workflows. Quanti-
436 fying internal reliability within this corpus would require re-annotating a subsample
437 with independent trained reviewers; however, agreement between Cochrane-trained
438 reviewers and non-specialist re-annotators would conflate task difficulty with exper-
439 tise asymmetry—the same confound that limits the interpretability of comparisons
440 between Cochrane and external reviewers [4]. The relevant population-level IRR
441 estimate is therefore taken from Hartling et al. [12], who measured agreement between
442 formally trained reviewer pairs on the same protocol: $\kappa = 0.79$ for D1 but only

443 $\kappa = 0.24\text{--}0.37$ for remaining domains, falling to $\kappa = 0.05\text{--}0.09$ between independent
 444 consensus pairs; independent Cochrane teams show similarly variable agreement [3].
 445 These figures establish that the reference standard itself carries substantial noise,
 446 placing a practical ceiling on any automated system’s agreement with published an-
 447 notations that is independent of model quality. Second, each model was queried once
 448 per study using provider-default generation parameters. This reflects the intended
 449 evaluation scope: zero-shot performance as models are available to practitioners,
 450 without artificial parameter tuning. Manipulating generation temperature would
 451 introduce heterogeneous conditions across model families, and strategies such as
 452 majority voting over repeated draws represent post-hoc optimisation outside the
 453 zero-shot benchmark design. As a consequence, differences between models separated
 454 by fewer than five percentage points should not be interpreted as definitive rank
 455 orderings in the absence of confidence intervals; these are discussed as directional
 456 trends. Third, all models were accessed in June 2026 via the OpenRouter API using
 457 the identifier strings reported in Table 2; providers may silently update the weights
 458 associated with a fixed identifier, so exact reproduction requires accessing the same
 459 model strings within the same inference window. Results should be interpreted as
 460 reflecting the checkpoint state active in June 2026, not any future revision of the
 461 same identifier. Fourth, the single-agent zero-shot design examines one specific slice
 462 of the capability space; few-shot prompting, retrieval augmentation, fine-tuning, and
 463 multi-agent architectures were not evaluated and may yield substantially different
 464 results. Fifth, training-data contamination cannot be excluded, but several struc-
 465 tural features of the evaluation design limit its plausibility as a primary driver of
 466 results. Memorisation-based performance would require a model to (i) have seen
 467 the particular one of the 127 distinct Cochrane 2024 reviews from which each study
 468 was drawn, (ii) correctly identify that the input Markdown document corresponds
 469 to a specific entry in that review’s RoB table, and (iii) reproduce the reviewer’s
 470 label rather than reason from the text. The Cochrane reviews sampled here were
 471 published in 2024; most included RCTs appear in paywalled journals, making full-text

472 ingestion at training time less likely for the majority of papers. Moreover, the empir-
473 ical pattern is inconsistent with memorisation at scale: the best individual model
474 (Claude Haiku 4.5) achieved a perfect 7/7 score on only 12 of 227 studies (5.3%), and
475 even the oracle across all eight models reached 7/7 on only 40 studies (17.6%). Both
476 figures are consistent with the rates expected under independent domain judgements
477 at the observed per-domain accuracy ($0.60^7 \approx 2.8\%$; inflated modestly by the low
478 intraclass correlation of $\hat{\rho} = 0.044$ to roughly 4–5% per model), leaving no systematic
479 excess attributable to retrieval. Targeted contamination probes (e.g. membership
480 inference or verbatim reproduction tests) were not conducted, as the opaque training
481 corpora of the evaluated models preclude rigorous analysis; this remains a caveat for
482 interpretation. Sixth, the Cochrane corpus used here comprises better-reported trials
483 than the broader trial literature, so these performance estimates may be optimistic
484 relative to unselected primary research.

485 Acknowledgements

486 Data and Code Availability

487 Conflicts of Interest

488 The authors declare no conflicts of interest.

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547 Supporting Information

548 S1 Prompt Used for Single-Agent RoB Assessment

549 The following prompt was used without modification across all eight models and all
550 227 studies. Each study was assessed in seven sequential inference calls—one per
551 RoB domain—with the full Markdown text of the study report included in every
552 call as the user message, followed by the domain identifier (e.g. Domain: D1). The
553 prompt text below was supplied as the system message.

554 Goal: Systematically assess the risk of bias in a research article using
555 Cochrane RoB 1.0. Evaluate bias domain, assign a judgment (Low, High,
556 or Unclear Risk), and justify strictly from the article evidence.

557

558 ---

559

560 Preliminary Considerations

561

562 Before assessing, identify:

563 1. Intervention Type: Pharmacological | Surgical/Procedural |

564 Non-pharmacological | Device

565 2. Primary Outcome Type: Objective (mortality, lab values) |

566 Subjective (pain, QoL)

567 3. Comparator: Placebo | Active comparator | Usual care | No intervention

568

569 Critical Context: Blinding expectations vary by intervention type.

570 For pharmacological interventions, lack of placebo is a serious flaw.

571 For surgical/non-pharmacological interventions where blinding is impossible,
572 focus on compensatory measures: blinding of outcome assessors and objective
573 endpoints.

574

575 ---

576

577 Bias Domains

578

579 1. Random Sequence Generation (Selection Bias)

580 Signalling Question: Was a truly random method used?

581 Low Risk: Random method adequately described (computer-generated,
582 random tables, coin toss, dice, shuffled envelopes,
583 minimization with random element)

584 High Risk: Non-random method (alternation, date of birth, hospital
585 record number, clinician judgment)

586 Unclear Risk: States "randomized" but no method details provided

587

588 2. Allocation Concealment (Selection Bias)

589 Signalling Question: Was the allocation sequence concealed from those
590 enrolling participants until assignment?

591 Low Risk: Adequate concealment (central/telephone randomization,
592 sequentially numbered opaque sealed envelopes, numbered
593 drug containers, independent pharmacy)

594 High Risk: Allocation not concealed or predictable (open list,
595 non-opaque envelopes, alternation)

596 Unclear Risk: Insufficient information to judge

597 Note: Allocation concealment (before randomization) is different from
598 blinding (after randomization).

599

600 3. Blinding of Participants and Personnel (Performance Bias)

601 Signalling Questions: Were participants/personnel aware of assignments?

602 If yes, could this affect outcomes?

603 Pharmacological: Double-blinding expected; no placebo implies higher
604 risk for subjective outcomes.

605 Surgical/Non-pharmacological: Blinding may be impossible and is not
606 automatically High Risk; check compensatory safeguards.

607 Objective outcomes: Less susceptible to performance bias.
608 Low Risk: Adequate blinding, OR blinding impossible with compensatory
609 measures, OR outcome is objective/hard endpoint
610 High Risk: No blinding + subjective outcome + no compensatory measures
611 Unclear Risk: Insufficient blinding information
612

613 4. Blinding of Outcome Assessors (Detection Bias)

614 Signalling Questions: Were assessors blinded? Is outcome objective or
615 subjective?

616 Low Risk: Assessors blinded, OR objective outcome unlikely to be
617 influenced by assignment knowledge

618 High Risk: Assessors not blinded + subjective outcome requiring
619 interpretation

620 Unclear Risk: Insufficient information

621 Note: For patient-reported outcomes, participant acts as assessor.
622

623 5. Incomplete Outcome Data (Attrition Bias)

624 Signalling Questions: Are data available for most participants? Are
625 missing-data reasons balanced/unrelated to outcome?

626 Low Risk: Minimal missingness (<5%), OR balanced missingness with
627 unrelated reasons, OR valid ITT/imputation

628 High Risk: High missingness (>10-20%), OR imbalance between groups,
629 OR outcome-related missingness, OR per-protocol exclusions

630 Unclear Risk: Missingness not adequately reported
631

632 6. Selective Reporting (Reporting Bias)

633 Signalling Questions: Is protocol/registry available? Were pre-specified
634 outcomes reported?

635 Low Risk: All pre-specified outcomes reported, OR protocol/registry
636 aligns with publication

637 High Risk: Selective non-reporting (methods outcomes missing in

```

638         results, changed primary outcome, significance-driven
639         reporting)
640     Unclear Risk: No protocol/registry and insufficient information
641
642 7. Other Bias
643     Signalling Questions: Are there other bias sources (early stopping,
644     baseline imbalance, conflicts, contamination)?
645     Low Risk:     No relevant additional bias detected
646     High Risk:    Important additional bias present
647     Unclear Risk: Insufficient information
648
649 ---
650
651 Key Principles
652 1. Evidence-based only: Base judgments on explicit information. Do not
653     assume. If evidence is missing, choose "Unclear Risk".
654 2. Cite evidence: Reference specific text or note absence of information.
655 3. Context matters: Apply blinding expectations appropriate to intervention
656     type.
657 4. Impossibility vs Negligence: Not blinding when impossible differs from
658     not blinding when feasible.
659 5. Outcome sensitivity: Subjective outcomes require stricter blinding than
660     objective ones.
661
662 ---
663
664 Return ONLY valid JSON with:
665 {
666     "judgement": "Low Risk" | "High Risk" | "Unclear Risk",
667     "justification": "short justification"
668 }

```

669 S2 Per-Domain Performance by Model

670 Tables S1 and S2 report accuracy (%) and Macro-F1 for each model in each of the
671 seven RoB domains.

672 S3 Stratification by Intervention Type

673 Table S3 reports Macro-F1 for each model within each intervention type. Figure S1
674 shows the same data as a heatmap.

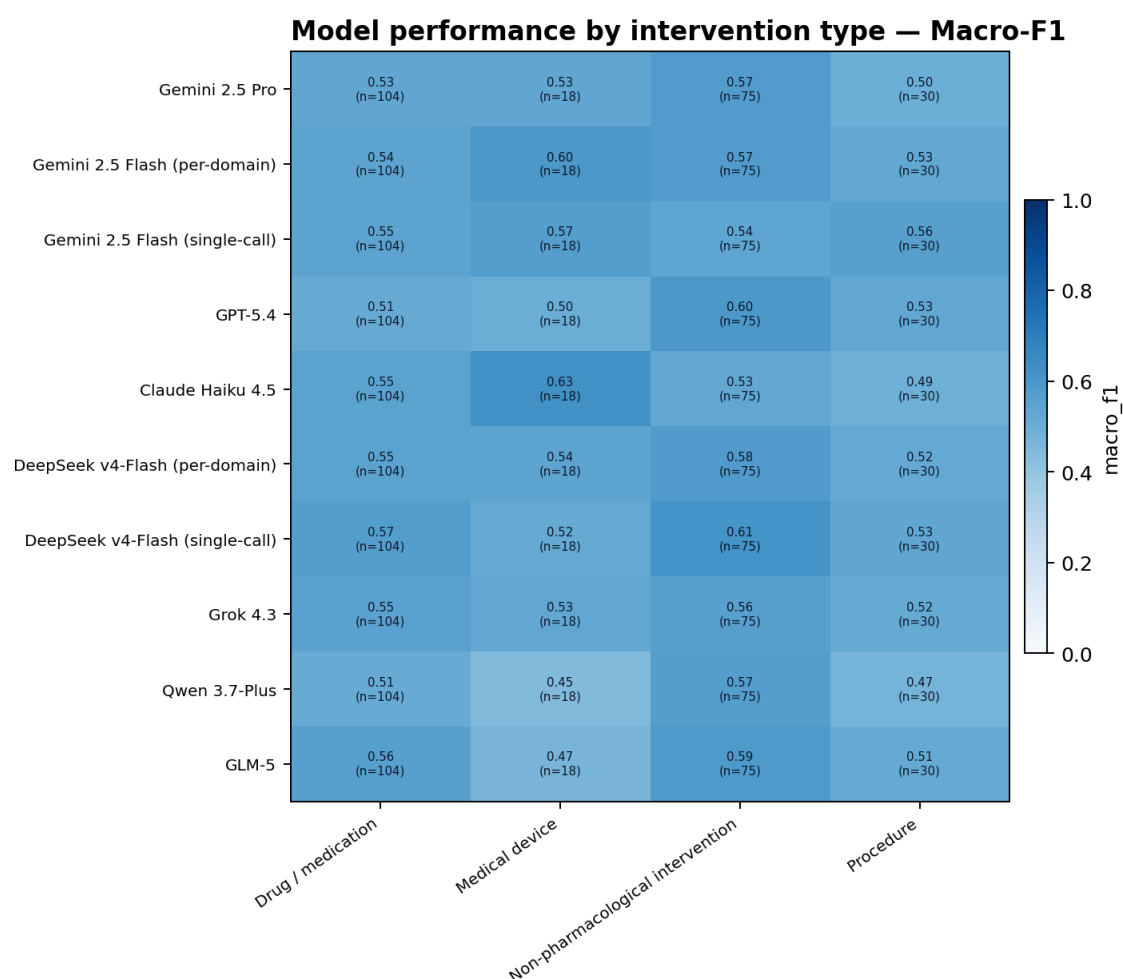


Figure S1: Macro-F1 heatmap by intervention type and model. Colour scale: 0–1.

675 S4 Stratification by Clinical Domain

676 Table S4 reports mean Macro-F1 (averaged across all eight models) for all 24 Cochrane
677 Review Group categories, ordered from highest to lowest mean F1. Domains with

Table S1: Per-domain accuracy (%) for all eight models ($n = 227$ cells per domain). Best value per domain in bold.

Model	D1	D2	D3	D4	D5	D6	D7
Claude Haiku 4.5	73.6	74.0	67.4	51.5	64.3	57.7	48.9
GLM-5	76.7	74.0	55.9	53.7	62.1	55.9	49.3
Qwen 3.7-Plus	77.5	74.9	49.8	52.0	59.9	56.4	55.5
Gemini 2.5 Flash	72.2	74.0	72.2	51.5	60.4	62.6	31.7
DeepSeek v4-Flash	74.0	70.0	67.0	57.7	54.6	55.9	42.7
Grok 4.3	75.8	70.0	59.5	54.6	58.6	60.8	41.4
GPT-5.4	75.8	70.0	64.3	52.0	51.5	55.9	34.4
Gemini 2.5 Pro	76.7	74.4	62.6	55.1	55.1	55.1	24.2

Table S2: Per-domain Macro-F1 for all eight models ($n = 227$ cells per domain). Best value per domain in bold.

Model	D1	D2	D3	D4	D5	D6	D7
Claude Haiku 4.5	0.572	0.618	0.527	0.425	0.394	0.387	0.269
GLM-5	0.632	0.657	0.462	0.479	0.514	0.443	0.345
Qwen 3.7-Plus	0.638	0.672	0.428	0.447	0.439	0.388	0.288
Gemini 2.5 Flash	0.549	0.663	0.626	0.411	0.412	0.438	0.280
DeepSeek v4-Flash	0.602	0.610	0.583	0.521	0.419	0.424	0.347
Grok 4.3	0.618	0.668	0.524	0.492	0.423	0.449	0.315
GPT-5.4	0.625	0.635	0.515	0.469	0.419	0.397	0.317
Gemini 2.5 Pro	0.621	0.671	0.481	0.478	0.413	0.475	0.195

Table S3: Macro-F1 stratified by intervention type. Numbers in parentheses are study counts (n_{studies}). Best value per row in bold.

Model	Drug / medication ($n = 104$)	Non-pharma- cological ($n = 75$)	Procedure ($n = 30$)	Medical device ($n = 18$)
Claude Haiku 4.5	0.548	0.530	0.494	0.627
GLM-5	0.559	0.590	0.514	0.471
Qwen 3.7-Plus	0.511	0.567	0.474	0.445
Gemini 2.5 Flash	0.540	0.574	0.529	0.596
DeepSeek v4-Flash	0.548	0.582	0.520	0.543
Grok 4.3	0.552	0.561	0.519	0.531
GPT-5.4	0.515	0.595	0.530	0.502
Gemini 2.5 Pro	0.533	0.575	0.498	0.534

fewer than five studies are included for completeness but should be interpreted with caution.

S5 Pairwise Model Complementarity

Table S5 reports pairwise complementarity across the eight models. Each cell (i, j) gives the percentage of the 1,589 study-domain cells in which model i produced an incorrect judgement and model j produced the correct one. High values indicate that model j recovers errors that model i misses, i.e. that j adds non-redundant information over i . The matrix is asymmetric: cell $(i, j) \neq$ cell (j, i) because the two models may have different overall accuracy.

S6 Confidence Interval Method Comparison

All confidence intervals reported in the main text were computed using study-clustered bootstrap (5,000 iterations, resampling the 227 studies with replacement). This section documents the comparison between bootstrap and two Wilson score interval variants for accuracy, and explains why the bootstrap approach was adopted for all metrics.

Table S4: Mean Macro-F1 across all eight models, stratified by clinical domain (24 Cochrane Review Groups, multi-label; study counts sum to more than 227). Ordered by descending mean F1.

Clinical domain	n_{studies}	Mean Macro-F1
Orthopaedics & trauma	7	0.680
Endocrine & metabolic	10	0.638
Rheumatology	5	0.637
Wounds	5	0.628
Dentistry & oral health	15	0.597
Lungs & airways	14	0.589
Pregnancy & childbirth	9	0.583
Neonatal	17	0.578
Complementary & alternative medicine	19	0.555
Child health	72	0.550
Gastroenterology & hepatology	21	0.538
Heart & circulation	45	0.536
Neurology	11	0.519
Cancer	21	0.494
Effective practice & health systems	4	0.487
Kidney disease	31	0.484
Mental health	11	0.481
Gynaecology	11	0.480
Insurance medicine	21	0.477
Blood disorders	6	0.476
Genetic disorders	9	0.473
Skin disorders	3	0.448
Reproductive & sexual health	8	0.437
Infectious disease	8	0.425

Wilson score interval

For a proportion p observed over n independent Bernoulli trials, the Wilson 95% CI is

$$\left(\frac{np + \frac{z^2}{2}}{n + z^2} \pm \frac{z \sqrt{np(1-p) + \frac{z^2}{4}}}{n + z^2} \right), \quad (\text{S1})$$

with $z = 1.96$. Applied to accuracy ($p \approx 0.60$), two natural choices for n exist: $n = 1,589$ (all annotated cells, assuming independence) or $n = 227$ (studies as the primary sampling unit).

Table S5: Pairwise complementarity matrix. Cell (i, j) = % of 1,589 cells where row model i is incorrect and column model j is correct. Row model abbreviations: Haiku = Claude Haiku 4.5; GLM-5 = GLM-5; Qwen = Qwen 3.7-Plus; Flash = Gemini 2.5 Flash; DSv4 = DeepSeek v4-Flash; Grok = Grok 4.3; GPT = GPT-5.4; Pro = Gemini 2.5 Pro.

	Haiku	GLM-5	Qwen	Flash	DSv4	Grok	GPT	Pro
Haiku	—	12.5	10.6	9.9	12.3	12.5	12.6	12.4
GLM-5	13.9	—	7.4	12.3	9.6	8.9	9.5	7.6
Qwen	12.3	7.6	—	11.7	10.3	7.6	9.6	9.4
Flash	11.7	12.8	11.9	—	11.3	10.2	9.4	9.1
DSv4	14.5	10.4	10.8	11.7	—	9.6	9.5	9.9
Grok	14.9	9.9	8.4	10.8	9.8	—	9.0	8.4
GPT	17.4	12.9	12.7	12.4	12.1	11.4	—	10.1
Pro	17.3	11.1	12.6	12.2	12.6	11.0	10.3	—

699 **Intraclass correlation and effective sample size**

700 The 1,589 cells are not independent: seven domain judgements are drawn from
701 the same study. Pearson’s one-way intraclass correlation coefficient (ICC) for the
702 correct/wrong binary indicator across the seven domains within each study was
703 estimated at $\hat{\rho} = 0.044$ (computed for Claude Haiku 4.5 as a representative model).
704 The low ICC indicates that whether a model correctly judges one domain bears little
705 information about its judgement on another domain of the same study, which is
706 consistent with the domain-specific nature of each RoB criterion.

707 Using the Kish (1965) design-effect formula, the effective sample size is

$$n_{\text{eff}} = \frac{n_{\text{cells}}}{1 + (k - 1)\hat{\rho}} = \frac{1,589}{1 + 6 \times 0.044} \approx 1,257, \quad (\text{S2})$$

708 where $k = 7$ domains per study.

709 **Numerical comparison**

710 Table S6 reports accuracy CIs for five models under three methods.

Table S6: Comparison of 95% confidence interval methods for overall accuracy. Width in percentage points (pp). Bootstrap: study-clustered, 5,000 iterations. Wilson n_{cells} : Equation (S1) with $n = 1,589$. Wilson n_{eff} : Equation (S1) with $n = 1,257$ (Kish formula). Wilson n_{studies} : Equation (S1) with $n = 227$.

Model	Accuracy	Wilson n_{cells} (width)	Wilson n_{eff} (width)	Bootstrap (width)
Claude Haiku 4.5	62.5%	[60.1, 64.8] (4.8)	[59.8, 65.1] (5.3)	[59.8, 65.2] (5.4)
Gemini 2.5 Flash	60.7%	[58.2, 63.0] (4.8)	[57.9, 63.4] (5.4)	[58.2, 63.1] (4.9)
DeepSeek v4-Flash	60.3%	[57.9, 62.7] (4.8)	[57.6, 63.0] (5.4)	[57.7, 62.9] (5.2)
GPT-5.4	57.7%	[55.3, 60.1] (4.9)	[54.9, 60.4] (5.4)	[55.1, 60.3] (5.2)
Gemini 2.5 Pro	57.6%	[55.1, 60.0] (4.9)	[54.8, 60.3] (5.4)	[55.1, 60.0] (4.9)
Wilson $n_{\text{studies}} = 227$		widths ≈ 12.5 – 12.8 pp (not shown; see text)		

711 Wilson with n_{cells} underestimates uncertainty by $\sim 10\%$ relative to bootstrap
712 (widths 4.8 pp vs. 5.1 pp). Wilson with n_{eff} virtually matches the bootstrap (widths
713 5.3 pp vs. 5.1 pp); the two methods agree to within 0.1 pp for every model. Wilson
714 with n_{studies} overestimates uncertainty by $\sim 2.5\times$ because it discards the domain-level
715 information and treats each study as a single binary observation.

716 Why bootstrap was adopted for all metrics

717 The numerical equivalence between Wilson n_{eff} and bootstrap for accuracy confirms
718 that the low ICC ($\hat{\rho} = 0.044$) makes the independence assumption nearly tenable.
719 Nevertheless, bootstrap was used uniformly for three reasons. First, Macro-F1
720 and Gwet’s AC1 are composite statistics with no closed-form variance expression;
721 bootstrap applies without modification. Second, the paired-bootstrap procedure used
722 for inference-mode Δ values (Table 4) jointly resamples both conditions, correctly
723 propagating correlation between them; Wilson intervals cannot be directly adapted
724 for paired differences of non-proportion statistics. Third, a single method for all
725 reported CIs ensures internal consistency and simplifies interpretation.